IMPLEMENTATION PLAN

Sathvani AVC — Artificial Vocal Cord

Version 1.0 | August 2026 | Confidential

1. DOCUMENT CONTROL

Field	Details
Document Title	Implementation Plan — Sathvani AVC
Version	1.0
Date	26 August 2026
Author	Sathvani Engineering & Operations Team
Classification	Confidential — Internal Use Only
Stakeholders	Founder/CEO, Engineering Lead, Clinical Advisor, Regulatory Consultant, Investors

2. EXECUTIVE SUMMARY

This implementation plan outlines the phased execution strategy to bring the Sathvani AVC (Artificial Vocal Cord) from current prototype status to commercial launch. The plan spans four major phases over 36 months: (1) Foundation & Data Collection, (2) Prototype & AI Training, (3) Clinical Validation & MVP Refinement, and (4) Regulatory Approval & Commercial Launch. Total seed funding requirement: ■30–50 lakhs for Phase 1–2; Series A target by end of 2027 for Phase 3–4 scale-up.

3. PHASE BREAKDOWN

PHASE 1: FOUNDATION & DATA COLLECTION (Q3–Q4 2025)

Duration: 6 months **| Budget:** ■8–12L **| Status:** Partially Completed

Objectives:

- Finalize WiFi+UDP streaming pipeline from ESP32-S3 to cloud
- Establish WAV data collection protocol with timestamp synchronization
- Begin AI training on available research dataset (48 phoneme instances, 6 dialogues)
- Complete hardware architecture design review and BOM freeze
- File provisional patent → initiate complete specification drafting

Deliverables:

- Working ESP32-S3 firmware with 4-sensor acquisition
- Cloud ingestion endpoint (v0.1) accepting UDP streams
- Baseline CNN-BiLSTM model trained on research data (100% accuracy on clean data)

• Provisional patent filed (already completed per pitch deck)

Key Risks:

• Research dataset is tiny (48 instances) — model may not generalize. Mitigation: begin recruiting laryngectomy patients for early data collection.

• Pressure sensor driver untested. Mitigation: bench test with calibrated pressure source.

PHASE 2: PROTOTYPE ASSEMBLY & AI MODEL V1 (Q1–Q2 2026)

Duration: 6 months | **Budget:** ■15–20L | **Status:** Upcoming

Objectives:

• Assemble first integrated wearable prototype (mask form factor)

• Train AI model v1 on expanded dataset (target: 500+ phoneme instances from 20+ speakers)

• Complete patent specification filing

• Design AIIMS Bibinagar pilot protocol and submit for IRB approval

• Build companion mobile app (MVP: Android, BLE pairing, audio playback)

Deliverables:

• 3–5 working prototype units for internal testing

• AI model v1 with >80% phoneme accuracy on held-out speaker data

• Complete patent specification filed (June 2026 target)

• IRB protocol approved at AIIMS

• Android companion app v0.1 (pairing + playback)

Milestones:

Milestone	Target Date	Owner	Success Criteria
M2.1	15 Jan 2026	Hardware Lead	Prototype PCB assembled and powered on
M2.2	28 Feb 2026	ML Lead	Model v1 trained; validation accuracy >80%
M2.3	31 Mar 2026	Regulatory	Complete patent spec filed with IPO
M2.4	30 Apr 2026	Clinical Advisor	IRB protocol approved at AIIMS
M2.5	31 May 2026	Software Lead	Android app v0.1 on Play Store (internal testing)
M2.6	30 Jun 2026	Founder/CEO	Seed round closed (■30–50L) or bridge financing secured

PHASE 3: CLINICAL VALIDATION & MVP REFINEMENT (Q3–Q4 2026)

Duration: 6 months | **Budget:** ■25–35L | **Status:** Upcoming

Objectives:

- Execute 10-patient clinical pilot at AIIMS Bibinagar
- Collect real patient data (laryngectomy + vocal cord paralysis) for model retraining
- Refine prototype into MVP based on clinical feedback (comfort, signal quality, usability)
- Achieve >85% phoneme accuracy on real patient data
- Initiate CDSCO manufacturing license application
- Build clinician dashboard and analytics platform

Deliverables:

- Clinical pilot report with IRB sign-off
- AI model v2 retrained on real patient data (>85% accuracy)
- MVP device (10 units) ready for expanded pilot
- CDSCO application submitted
- Clinician dashboard v1.0

Milestones:

Milestone	Target Date	Owner	Success Criteria
M3.1	15 Aug 2026	Clinical Advisor	First patient enrolled in pilot
M3.2	30 Sep 2026	ML Lead	Real patient dataset >200 phoneme instances
M3.3	31 Oct 2026	Hardware Lead	MVP design freeze based on patient feedback
M3.4	30 Nov 2026	ML Lead	Model v2 accuracy >85% on held-out patient data
M3.5	15 Dec 2026	Regulatory	CDSCO application dossier submitted
M3.6	31 Dec 2026	Founder/CEO	Pilot report published; Series A pitch deck ready

PHASE 4: REGULATORY APPROVAL & COMMERCIAL LAUNCH (2027–2028)

Duration: 18 months | **Budget:** ■1.5–2.5Cr (Series A) | **Status:** Upcoming

Objectives:

- Secure FDA 510(k) or De Novo clearance and CDSCO manufacturing license
- Launch hospital B2B sales via AIC-CCMB network and ENT surgeon partnerships
- Launch B2C D2C portal for direct patient access
- Scale manufacturing to 2,300+ units (Year 1 target)
- Build SaaS subscription layer (voice model updates, cloud processing)
- Expand to multi-language (Hindi, Telugu, Tamil, Kannada)

Deliverables:

- FDA clearance + CDSCO manufacturing license
- Commercial device (final BOM cost <■25K; retail ■65K)
- B2B portal + B2C e-commerce platform live
- SaaS tier launched (■500–1,500/month)

• ISO 13485 QMS certification

Milestones:

Milestone	Target Date	Owner	Success Criteria
M4.1	30 Jun 2027	Regulatory	CDSCO manufacturing license obtained
M4.2	30 Sep 2027	Regulatory	FDA pre-submission meeting completed
M4.3	31 Dec 2027	Founder/CEO	Series A closed (■3–5Cr)
M4.4	31 Mar 2028	Operations	First 500 units manufactured and shipped
M4.5	30 Jun 2028	Sales	2,300+ units distributed; ■1.5Cr revenue run-rate
M4.6	31 Dec 2028	Product	Multi-language support (4+ Indian languages)

4. RESOURCE PLAN

4.1 Team Structure

Role	Headcount	Responsibility	Timing
Founder / CEO	1	Strategy, fundraising, clinical partnerships, investor relations	Full-time from Day 0
ML / AI Lead	1	Model architecture, training, evaluation, feature engineering	Full-time from Day 0
Embedded Engineer	1	ESP32-S3 firmware, sensor integration, PCB design	Full-time from Day 0
Cloud / Backend Engineer	1	API design, infrastructure, DevOps, security	Hire by M2.1 (Jan 2026)
Mobile App Developer	1	Android companion app, BLE integration, UI/UX	Hire by M2.1 (Jan 2026)
Clinical Advisor (Consultant)	1	Trial design, IRB, regulatory guidance, patient recruitment	Part-time from Day 0
Regulatory Consultant	1	CDSCO/FDA submissions, ISO 13485, QMS setup	Hire by M2.3 (Mar 2026)
Industrial Designer	0.5	Mask ergonomics, comfort, aesthetic design	Contract by M2.3 (Mar 2026)
Manufacturing Partner	—	PCBA, assembly, packaging, supply chain	Engage by M3.3 (Oct 2026)

4.2 Budget Allocation (Seed ■30–50L)

Category	Amount (■L)	% of Seed	Purpose
Hardware & Sensors	8–12	25%	PVDF film, MEMS mics, pressure/airflow sensors, ESP32-S3 modules
Cloud & AI Infrastructure	5–8	18%	GPU compute (AWS/GCP), storage, model training, API hosting
Clinical Pilot	5–7	15%	AIIMS collaboration, patient compensation, IRB fees, clinical monitorin
Patent & Legal	3–5	10%	Complete patent spec, trademark, regulatory consulting
Software Development	4–6	14%	Mobile app, cloud platform, clinician dashboard
Team Salaries (6 months)	8–10	22%	Core team compensation (founder + 2 engineers)
Miscellaneous / Buffer	2–3	6%	Travel, office, contingency

5. RISK REGISTER & CONTINGENCY

Risk	Impact	Likelihood	Contingency Plan	Trigger
AI model fails to generalize to real patient data	Critical	High	Pivot to hybrid approach: piezo + electrolarynx acoustic fusion; expand sensor stack to inclu	Model v2 accuracy <70% on pilot data
Clinical pilot delayed (patient recruitment)	High	Medium	Expand to multiple sites (NIMHANS, CMC Vellore); offer patient [illegible] partner with laryn	<9 patients enrolled by M3.1 + 30 days
CDSCO rejects initial application	High	Low	Engage third-party regulatory consultant for remediation; consider Class A self-certification	Rejection letter received
Key team member departure	Medium	Medium	Maintain detailed documentation (this TRD); cross-train team members; offer equity vesting	30-day notice given
Seed funding insufficient	High	Medium	Apply to BIRAC, DST, NIDHI-PRAYAS grants; approach angel investors in medtech; reduc	<50% of seed target by Dec 2025
Supply chain disruption (sensors)	Medium	Low	Dual-source all critical components; maintain 3-month inventory buffer; identify local alterna	Lead time >8 weeks

6. GOVERNANCE & REPORTING

Weekly: Engineering standup (30 min) — blockers, sprint progress, bug triage

Bi-weekly: Product review — demo new features, review metrics, adjust priorities

Monthly: Board/Advisor update — financials, milestones, risks, fundraising status

Quarterly: Strategic review — OKRs, market feedback, competitive landscape, pivot decisions

Ad-hoc: Clinical safety review — any adverse events in pilot trigger immediate review within 24 hours

7. SUCCESS CRITERIA BY PHASE

Phase	Technical Success	Business Success	Clinical Success
Phase 1 (2025)	4-sensor acquisition working; UDP pipeline stable; baseline model trained	Provisional patent filed; seed pitch deck validated	—
Phase 2 (Q1-Q2 2026)	Model v1 >80% accuracy; Android app v0.1; 3–5 prototypes	Seed round closed (■30–50L); complete patent filed	IRB protocol approved
Phase 3 (Q3-Q4 2026)	Model v2 >85% on patient data; MVP design freeze; clinical dashboard	Series A pitch ready; CDSCO app submitted	10-patient pilot completed; safety profile
Phase 4 (2027–2028)	Commercial firmware v1.0; multi-language TTS; edge quantization R&D	2,300+ units sold; ■1.5Cr revenue; SaaS launched	FDA/CDSCO approved; real-world evide

8. DEPENDENCIES & CRITICAL PATH

Critical Path (longest duration to commercial launch):

1. **Sensor signal validation on patients** (Phase 1–2) → Gates all AI model reliability
2. **AI model v2 training on real data** (Phase 3) → Gates clinical efficacy claim
3. **10-patient pilot completion** (Phase 3) → Gates regulatory submission and investor confidence
4. **CDSCO manufacturing license** (Phase 4) → Gates commercial sales in India
5. **FDA clearance** (Phase 4) → Gates US market entry and global credibility

Parallel Tracks (non-critical but important):
• Mobile app development can proceed in parallel with hardware
• Patent prosecution runs in parallel with clinical work
• SaaS platform development can begin after MVP design freeze

9. APPENDICES

Appendix A: Gantt Chart Summary — See project management tool (Notion/Jira) for detailed task-level tracking.

Appendix B: Vendor Contact List — Sensor suppliers, PCB manufacturers, contract manufacturers (CM), cloud providers.

Appendix C: Clinical Protocol Draft — AIIMS Bibinagar 10-patient pilot: inclusion/exclusion criteria, endpoints, safety monitoring plan.

Appendix D: Competitive Benchmarking — Silent speech interface research (MIT AlterEgo, EMG-based systems) as technical reference class.

— End of Implementation Plan —